KERALA DIPLOMA CURRICULUM REVISION 2026

Course 2072 | II | As specified in the official PDF | Source-grounded

Fundamentals of Chemical Engineering

Engineering technicians. operations and unit processes

OFFICIAL SOURCE DECLARATION

How this lesson was prepared

The supplied official SITTTR syllabus PDF for Course 2072 is the authoritative source for course information, objectives, course outcomes, CO–PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

Source file: 2072.pdf from the uploaded official SITTTR syllabus archive. **Course code and title:** preserved from the source.

COURSE DASHBOARD

Official course information

Course information

Item	Official information
Programme	Diploma Programme
Course code	2072
Course title	Fundamentals of Chemical Engineering
Semester	II
Course category	As specified in the official PDF

Item	Official information
Periods per week	As specified in the official PDF
Periods per semester	As specified in the official PDF
Credits	As specified in the official PDF
Prerequisite	Topic Course code Course Title Semester Fundamentals of Physics 2002B Engineering Physics for Applied Electrical Technology and Computing I Fundamentals of Mathematics 1002 Fundamentals of Engineering Mathematics I Course Outcome Blooms Taxonomy CO Course Outcome Level Summarise the historical background of chemical engineering, its scope, and the duties of a chemical CO1 Understand engineering technician, while applying the basic calculations in unit conversions. Summarise the fundamentals of unit operations, unit processes, momentum, heat, and mass transfer CO2 Understand operations. Outline the basic principles and terminology of electrical and electronics engineering relevant to CO3 Understand chemical processes. CO4 Summarise the fundamental concepts of mechanics applicable to chemical engineering. Understand CO-PO mapping COURSE ARTICULATION MATRIX PROGRAM OUTCOMES
Assessment	Use the official assessment section reproduced below

Modules

4 official module sections detected.

Topic entries

19 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

STUDY METHOD

How to use this lesson

First pass

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES

Objectives, outcomes and mapping

Course objectives

- Engineering technicians. operations and unit processes

Course outcomes

CO	Official outcome	Study level
CO1	3 - - - - - - - - - 3	See official syllabus
CO2	2 - - - - - - - - - 3	See official syllabus
CO3	2 - - - - - - - - - 3	See official syllabus
CO4	2 - - - - - - - - - 3 Total 9 - - - - - - - - - 12 Correlation Level 2.33 - - - - - - - - - 3 Legends: 1 - Low; 2 - Medium; 3 - High; No Correlation - “-” Teaching and Assessment Scheme Diploma Curriculum Revision 2026 2072 Page #1 Examination Scheme Teaching Scheme/Week Self-learning Total Credits Theory Practical (SS)/Week Hours/Semester C Total L T P S CIE ESE Total CIE ESE Total 4 0 0 0 6 60 4 40 60 100 0 0 0 100 L: Lecture (One unit is of one-hour duration with one credit), T: Tutorial (One unit is of one-hour duration with one credit), P: Practical (One unit is of one-hour duration with 0.5 credit), S: Project (One unit is of	See official syllabus

CO	Official outcome	Study level
	one-hour duration with 0.5 credit) SS: Self-Learning, CIE: Continuous Internal Evaluation, ESE: End Semester Examination Syllabus - Major Topics Instructional Module Title Major Topics Hours L T Historical background and scope of Chemical Engineering - Duties of a Chemical Engineering Technician - Differentiate the duties of a chemical engineering Introduction to technician and a chemist.	

CO–PO mapping evidence

Row	Extracted official mapping
CO1	CO1 3 - - - - - - - - - 3
CO2	CO2 2 - - - - - - - - - 3
CO3	CO3 2 - - - - - - - - - 3
CO4	CO4 2 - - - - - - - - - 3

COVERAGE AUDIT

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

Module	Official heading	Topic entries	Hours
Module 1	15 0	4	As specified
Module 2	15 0	3	As specified
Module 3	15 0	5	As specified
Module 4	Basics in Mechanics 15 0	7	As specified

MODULE 1

15 0

This module is presented from the official syllabus scope for Fundamentals of Chemical Engineering. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: 15 0
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

chemical industries Modern Chemical industries – Batch and continuous processes – transition from

chemical industries Modern Chemical industries – Batch and continuous processes – transition from is an official topic in Module 1 of Fundamentals of Chemical Engineering. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify chemical industries Modern Chemical industries – Batch and continuous processes – transition from using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, chemical industries modern chemical industries – batch and continuous processes – transition from should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What chemical industries Modern Chemical industries – Batch and continuous processes – transition from means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ chemical industries Modern Chemical industries – Batch and continuous processes – transition from □□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□.

batch to continuous processes.

batch to continuous processes. is an official topic in Module 1 of Fundamentals of Chemical Engineering. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify batch to continuous processes. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, batch to continuous processes. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What batch to continuous processes. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-[illegible] batch to continuous processes. [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Units and conversions – basic calculations.

Units and conversions – basic calculations. is an official topic in Module 1 of Fundamentals of Chemical Engineering. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Units and conversions – basic calculations. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, units and conversions – basic calculations. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Units and conversions – basic calculations. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-[illegible] Units and conversions – basic calculations. [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Unit operations and Fundamentals of momentum, heat and mass transfer – Unit Operations – Unit

Unit operations and Fundamentals of momentum, heat and mass transfer – Unit Operations – Unit is an official topic in Module 1 of Fundamentals of Chemical Engineering. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Unit operations and Fundamentals of momentum, heat and mass transfer – Unit Operations – Unit using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, unit operations and fundamentals of momentum, heat and mass transfer – unit operations – unit should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Unit operations and Fundamentals of momentum, heat and mass transfer – Unit Operations – Unit means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ Unit operations and Fundamentals of momentum, heat and mass transfer – Unit Operations – Unit □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
15           0
              chemical industries        Modern Chemical industries – Batch and
continuous processes – transition from
                                         batch to continuous processes.
                                         Units and conversions – basic
calculations.
              Unit operations and        Fundamentals of momentum, heat and mass
transfer – Unit Operations – Unit
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 2

15 0

This module is presented from the official syllabus scope for Fundamentals of Chemical Engineering. Use the topic cards for learning and the source transcript for exact coverage.

As specified **CO-linked**

official hours | see outcomes

Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: 15 0
- Official duration: As specified
- Official topic entries captured: 3
- The full source excerpt is preserved below for coverage checking.

unit processes Processes.

unit processes Processes. is an official topic in Module 2 of Fundamentals of Chemical Engineering. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify unit processes Processes. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, unit processes processes. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What unit processes Processes. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ unit processes Processes. □□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□□.

Current – Voltage – Resistance – Power – Ohm’s law – Laws of resistance –

Current – Voltage – Resistance – Power – Ohm’s law – Laws of resistance – is an official topic in Module 2 of Fundamentals of Chemical Engineering. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Current – Voltage – Resistance – Power – Ohm’s law – Laws of resistance – using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, current – voltage – resistance – power – ohm’s law – laws of resistance – should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Current – Voltage – Resistance – Power – Ohm’s law – Laws of resistance – means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 2-□□ Current – Voltage – Resistance – Power – Ohm’s law – Laws of resistance – □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

Basic Electrical and laws of electromagnetic induction – Generation of AC voltage – Transformers –

Basic Electrical and laws of electromagnetic induction – Generation of AC voltage – Transformers – is an official topic in Module 2 of Fundamentals of Chemical Engineering. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Basic Electrical and laws of electromagnetic induction – Generation of AC voltage – Transformers – using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, basic electrical and laws of electromagnetic induction – generation of ac voltage – transformers – should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Basic Electrical and laws of electromagnetic induction – Generation of AC voltage – Transformers – means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ Basic Electrical and laws of electromagnetic induction – Generation of AC voltage – Transformers – □□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
15          0
             unit processes             Processes.
                                  Current – Voltage – Resistance – Power – Ohm’s
law – Laws of resistance –
                                  Combination of resistances – series, parallel –
DC circuit – AC circuit – Faraday’s
           Basic Electrical and    laws of electromagnetic induction – Generation
of AC voltage – Transformers –
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 3

15 0

This module is presented from the official syllabus scope for Fundamentals of Chemical Engineering. Use the topic cards for learning and the source transcript for exact coverage.

As specified **CO-linked**

official hours | see outcomes

Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: 15 0
- Official duration: As specified
- Official topic entries captured: 5
- The full source excerpt is preserved below for coverage checking.

Electronics Engineering Electric motors – Applications in chemical plants.

Electronics Engineering Electric motors – Applications in chemical plants. is an official topic in Module 3 of Fundamentals of Chemical Engineering. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Electronics Engineering Electric motors – Applications in chemical plants. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, electronics engineering electric motors – applications in chemical plants. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Electronics Engineering Electric motors – Applications in chemical plants. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 3-[illegible] Electronics Engineering Electric motors – Applications in chemical plants. [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Electronic components – Resistors – Capacitors – Inductors – Diodes – Applications

Electronic components – Resistors – Capacitors – Inductors – Diodes – Applications is an official topic in Module 3 of Fundamentals of Chemical Engineering. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Electronic components – Resistors – Capacitors – Inductors – Diodes – Applications using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, electronic components – resistors – capacitors – inductors – diodes – applications should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Electronic components – Resistors – Capacitors – Inductors – Diodes – Applications means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-[illegible] Electronic components – Resistors – Capacitors – Inductors – Diodes – Applications [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

in chemical industries.

in chemical industries. is an official topic in Module 3 of Fundamentals of Chemical Engineering. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify in chemical industries. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, in chemical industries. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What in chemical industries. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-[illegible] in chemical industries. [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Basics of mechanics – Statics – Dynamics – force – Resolution of force – Friction

Basics of mechanics – Statics – Dynamics – force – Resolution of force – Friction is an official topic in Module 3 of Fundamentals of Chemical Engineering. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Basics of mechanics – Statics – Dynamics – force – Resolution of force – Friction using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, basics of mechanics – statics – dynamics – force – resolution of force – friction should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Basics of mechanics – Statics – Dynamics – force – Resolution of force – Friction means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ Basics of mechanics – Statics – Dynamics – force – Resolution of force – Friction □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

types and laws of friction – Stresses and strain — Types of stresses – Standard

types and laws of friction – Stresses and strain — Types of stresses – Standard is an official topic in Module 3 of Fundamentals of Chemical Engineering. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify types and laws of friction – Stresses and strain — Types of stresses – Standard using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, types and laws of friction – stresses and strain — types of stresses – standard should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What types and laws of friction – Stresses and strain — Types of stresses – Standard means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 3-□□ types and laws of friction – Stresses and strain — Types of stresses – Standard □□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
15           0
           Electronics Engineering Electric motors – Applications in chemical
plants.
                                   Electronic components – Resistors – Capacitors –
Inductors – Diodes – Applications
                                   in chemical industries.
                                         Basics of mechanics – Statics – Dynamics –
force – Resolution of force – Friction
                                         types and laws of friction – Stresses and
strain – Types of stresses – Standard
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 4

Basics in Mechanics 15 0

This module is presented from the official syllabus scope for Fundamentals of Chemical Engineering. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Basics in Mechanics 15 0
- Official duration: As specified
- Official topic entries captured: 7
- The full source excerpt is preserved below for coverage checking.

Basics in Mechanics 15 0

Basics in Mechanics 15 0 is an official topic in Module 4 of Fundamentals of Chemical Engineering. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Basics in Mechanics 15 0 using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, basics in mechanics 15 0 should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Basics in Mechanics 15 0 means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ Basics in Mechanics 15 0 □□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□.

stress-strain curve – Mechanical properties of materials – Elasticity, stiffness,

stress-strain curve – Mechanical properties of materials – Elasticity, stiffness, is an official topic in Module 4 of Fundamentals of Chemical Engineering. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify stress-strain curve – Mechanical properties of materials – Elasticity, stiffness, using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, stress-strain curve – mechanical properties of materials – elasticity, stiffness, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What stress-strain curve – Mechanical properties of materials – Elasticity, stiffness, means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 4-□□ stress-strain curve – Mechanical properties of materials – Elasticity, stiffness, □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

plasticity, toughness, brittleness, ductility, malleability and hardness.

plasticity, toughness, brittleness, ductility, malleability and hardness. is an official topic in Module 4 of Fundamentals of Chemical Engineering. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify plasticity, toughness, brittleness, ductility, malleability and hardness. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, plasticity, toughness, brittleness, ductility, malleability and hardness. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What plasticity, toughness, brittleness, ductility, malleability and hardness. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 4-[illegible] plasticity, toughness, brittleness, ductility, malleability and hardness. [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Detailed Syllabus

Detailed Syllabus is an official topic in Module 4 of Fundamentals of Chemical Engineering. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Detailed Syllabus using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, detailed syllabus should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Detailed Syllabus means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-[illegible] Detailed Syllabus [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Mode of

Mode of is an official topic in Module 4 of Fundamentals of Chemical Engineering. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Mode of using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, mode of should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Mode of means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-[illegible] Mode of [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Mapped Instructional

Mapped Instructional is an official topic in Module 4 of Fundamentals of Chemical Engineering. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Mapped Instructional using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, mapped instructional should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Mapped Instructional means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ Mapped Instructional □□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□□.

Subtopic Student Learning Outcome delivery Taxonomy Level

Subtopic Student Learning Outcome delivery Taxonomy Level is an official topic in Module 4 of Fundamentals of Chemical Engineering. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Subtopic Student Learning Outcome delivery Taxonomy Level using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, subtopic student learning outcome delivery taxonomy level should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Subtopic Student Learning Outcome delivery Taxonomy Level means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 4-□□ Subtopic Student Learning Outcome delivery Taxonomy Level □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
Basics in Mechanics
15           0
                                          stress-strain curve – Mechanical
properties of materials – Elasticity, stiffness,
                                          plasticity, toughness, brittleness,
ductility, malleability and hardness.

Detailed Syllabus

Mode of

Mapped                              Instructional
             Subtopic                           Student Learning Outcome
delivery              Taxonomy Level

CO                                    Hours

(L/T)
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

RAPID REFERENCE

Concept and formula bank

Answer structure

```
Definition → Principle → Components/steps → Application → Conclusion
```

Use this sequence for descriptive questions when the topic does not require a numerical formula.

Practical record

```
Aim → Apparatus → Procedure → Observation → Result → Safety
```

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION

Diagram and source-transcript library

Module 1 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING

Activities from the syllabus

Use the extracted official activity wording as the record of required self-learning work.

- for Self-Learning
- Week Self-Learning description Hours Module I
- Prepare notes on the scope of chemical engineering. Visit the websites of various industries in India to learn about their
- major products. 6
- Prepare notes to identify the duties of a chemical engineer
- Visit the NPTEL courses on important steps in process development to understand the chemical process development.
- Make a comparison table of Chemical Engineer/Technician/Chemist duties.
- Sketch batch vs continuous process flow diagrams.
- Refer to the textbook to identify the different unit systems
- Recall the mole concepts and definitions from the basic chemistry textbooks.
- Solve simple numericals based on molecular weights, equivalent weights, and the number of moles of the mixture.
- Create a Chemical Engineering timeline chart (1850-present). Prepare study notes for module I Module II
- Refer to textbooks on the fundamentals of chemical engineering to understand the basics of fluid mechanics, heat transfer,
- and mass transfer. 6
- Prepare study notes on Fluid Mechanics, Heat Transfer, and Mass Transfer - Basic Concepts
- Refer to text book on the concept of unit operations and unit processes.
- Prepare notes for various unit operations taught that week
- Refer to text book on various unit operations.
- Prepare notes for various unit operations taught in that week. 6
- Draw simple blocks of unit operations showing inputs and outputs

EXAMINATION PREPARATION

Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

```
– Theory
                                                 CA1                     CA2
CA3                   CA4                   CA5             CA6               ESE

       Mode           Attendance             Self-learning       Self-learning
Self-learning        Self-learning            Series      Series
Written
                                              activities          activities
activities           activities             Exam        Exam                      Exam
                                             (Module 1)          (Module 2)
(Module 3)           (Module 4)           (2 modules) (2 modules)
(theory)
   Duration
2 hours           2 hours          2.5 hours
  Exam mark                                         15                     15
15                     15                  50               50                 60
 Converted to
20               20
       Marks                5                                                          15
20                         60
       Total
40                                                                                  60
   Tentative              End of
End of
                                             By 4th week          By 7th week
By 11th week         By 14th week           8th week         15th week
   schedule               semester
```

Practice-paper notice: The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

Module 1

Define or explain chemical industries Modern Chemical industries – Batch and continuous processes – transition from. (3 marks)

Answer: Begin with the standard definition or identity of *chemical industries Modern Chemical industries – Batch and continuous processes – transition from*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain batch to continuous processes.. (3 marks)

Answer: Begin with the standard definition or identity of *batch to continuous processes.*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Units and conversions – basic calculations.. (3 marks)

Answer: Begin with the standard definition or identity of *Units and conversions – basic calculations.*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Unit operations and Fundamentals of momentum, heat and mass transfer – Unit Operations – Unit. (3 marks)

Answer: Begin with the standard definition or identity of *Unit operations and Fundamentals of momentum, heat and mass transfer – Unit Operations – Unit*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Module 2

Define or explain unit processes Processes.. (3 marks)

Answer: Begin with the standard definition or identity of *unit processes Processes.*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Current – Voltage – Resistance – Power – Ohm’s law – Laws of resistance –. (3 marks)

Answer: Begin with the standard definition or identity of *Current – Voltage – Resistance – Power – Ohm’s law – Laws of resistance* –. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Basic Electrical and laws of electromagnetic induction – Generation of AC voltage – Transformers –. (3 marks)

Answer: Begin with the standard definition or identity of *Basic Electrical and laws of electromagnetic induction – Generation of AC voltage – Transformers* –. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps,

□□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Module 3

Define or explain Electronics Engineering Electric motors – Applications in chemical plants.. (3 marks)

Answer: Begin with the standard definition or identity of *Electronics Engineering Electric motors – Applications in chemical plants.*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Electronic components – Resistors – Capacitors – Inductors – Diodes – Applications. (3 marks)

Answer: Begin with the standard definition or identity of *Electronic components – Resistors – Capacitors – Inductors – Diodes – Applications*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain in chemical industries.. (3 marks)

Answer: Begin with the standard definition or identity of *in chemical industries.*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Basics of mechanics – Statics – Dynamics – force – Resolution of force – Friction. (3 marks)

Answer: Begin with the standard definition or identity of *Basics of mechanics – Statics – Dynamics – force – Resolution of force – Friction*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Module 4

Define or explain Basics in Mechanics 15 0. (3 marks)

Answer: Begin with the standard definition or identity of *Basics in Mechanics 15 0*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain stress-strain curve – Mechanical properties of materials – Elasticity, stiffness,. (3 marks)

Answer: Begin with the standard definition or identity of *stress-strain curve – Mechanical properties of materials – Elasticity, stiffness,*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain plasticity, toughness, brittleness, ductility, malleability and hardness.. (3 marks)

Answer: Begin with the standard definition or identity of *plasticity, toughness, brittleness, ductility, malleability and hardness.*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□ □□□□□□□□□□ □□□□□□□.

Define or explain Detailed Syllabus. (3 marks)

Answer: Begin with the standard definition or identity of *Detailed Syllabus*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□ □□□□□□□□□□ □□□□□□□.

PRACTICE ONLY

Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

Module 1

1-mark definition: State the meaning of **chemical industries Modern Chemical industries – Batch and continuous processes – transition from**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 2

1-mark definition: State the meaning of **unit processes Processes.**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 3

1-mark definition: State the meaning of **Electronics Engineering Electric motors – Applications in chemical plants.**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 4

1-mark definition: State the meaning of **Basics in Mechanics 15 0**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION

Exam checklist

Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

Glossary

Study glossary

Term	Meaning in this lesson
Course Outcome	A capability the student should demonstrate after completing the course.
Module	An official syllabus unit with defined topics and hours.
CIE	Continuous Internal Evaluation.
ESE	End Semester Examination.
Source transcript	A preserved extract of the official syllabus used for coverage checking.

SOURCE-SUPPORTED REFERENCES

References and web resources

References listed in the official PDF

References are included in the official source PDF.

Web resources listed in the official PDF

- URL Modules Covered
- https://www.youtube.com/watch?v=i7t-MkFi6DU I

- https://nptel.ac.in/courses/108105112 III

IN-PAGE SEARCH

Search this lesson

Prepared from the supplied official SITTTR syllabus PDF for Course 2072. Verify institutional updates against the current official curriculum.